

Block 2

MBA Sustainability Strategy & AI · 15 Slides

Why Strategic Forecasting Matters

Strategic decisions increasingly involve long time horizons and high uncertainty. Companies investing in infrastructure, energy systems, supply chains, or new technologies must often make decisions whose consequences will unfold over decades.

Traditional Planning Limits

Traditional methods assume a stable environment, but modern systems are shaped by rapid technological innovation, geopolitical instability, and environmental change.

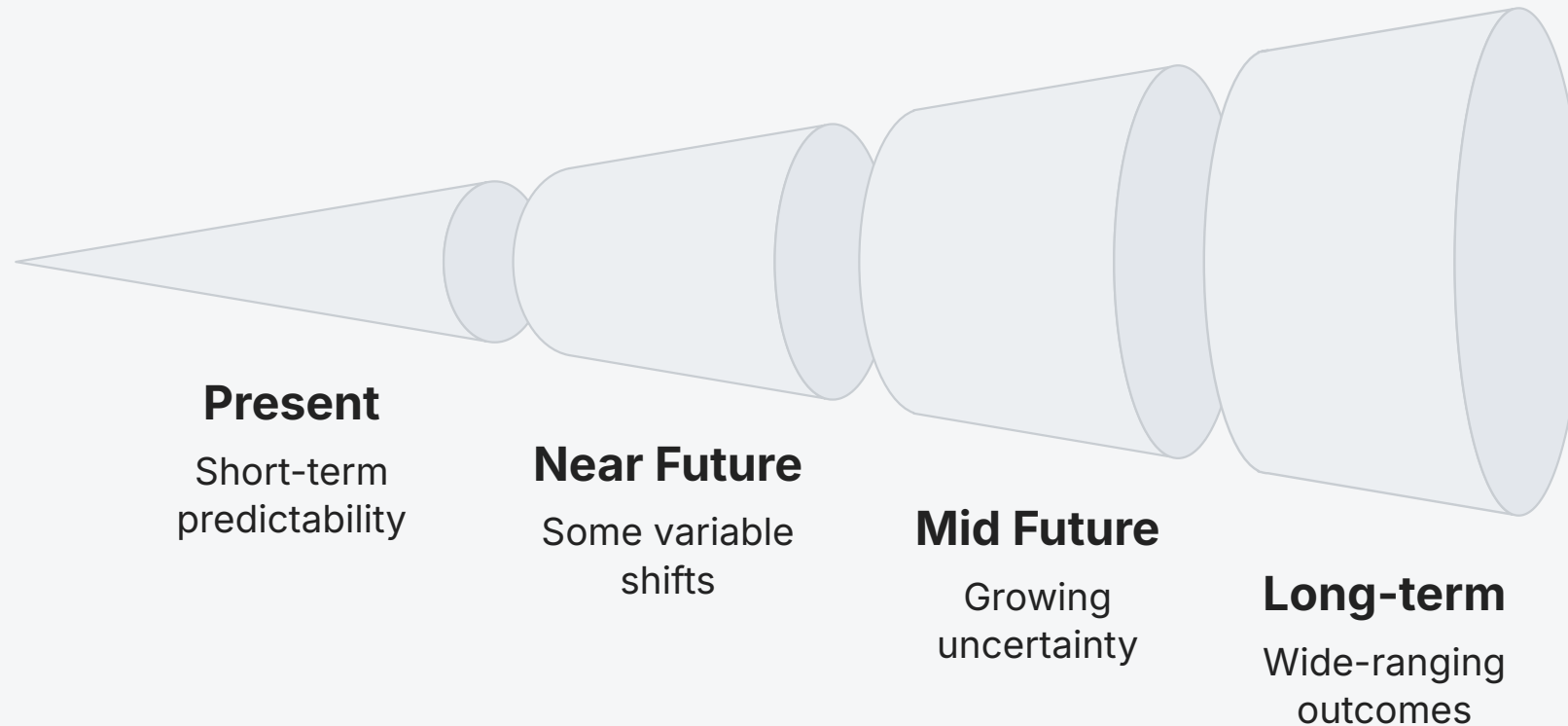
Exploring Multiple Futures

Rather than predicting a single future, strategic forecasting explores multiple plausible futures to prepare organizations for uncertainty.

Improved Decision-Making

This approach helps organizations design strategies that remain effective under different conditions.

The Cone of Uncertainty



- Short-term: Many variables remain predictable
- Long-term: Technology, politics, and environment create wide-ranging outcomes
- Strategic forecasting considers multiple possible futures rather than a single prediction

Types of Strategic Forecasting Tools

Organizations use several complementary methods to explore the future.



Trend Analysis

Identifying long-term patterns in technology, demographics, economics, or environmental systems.



Scenario Planning

Constructing alternative future worlds based on key uncertainties.



Systems Analysis

Understanding interactions between stakeholders, resources, and infrastructure.

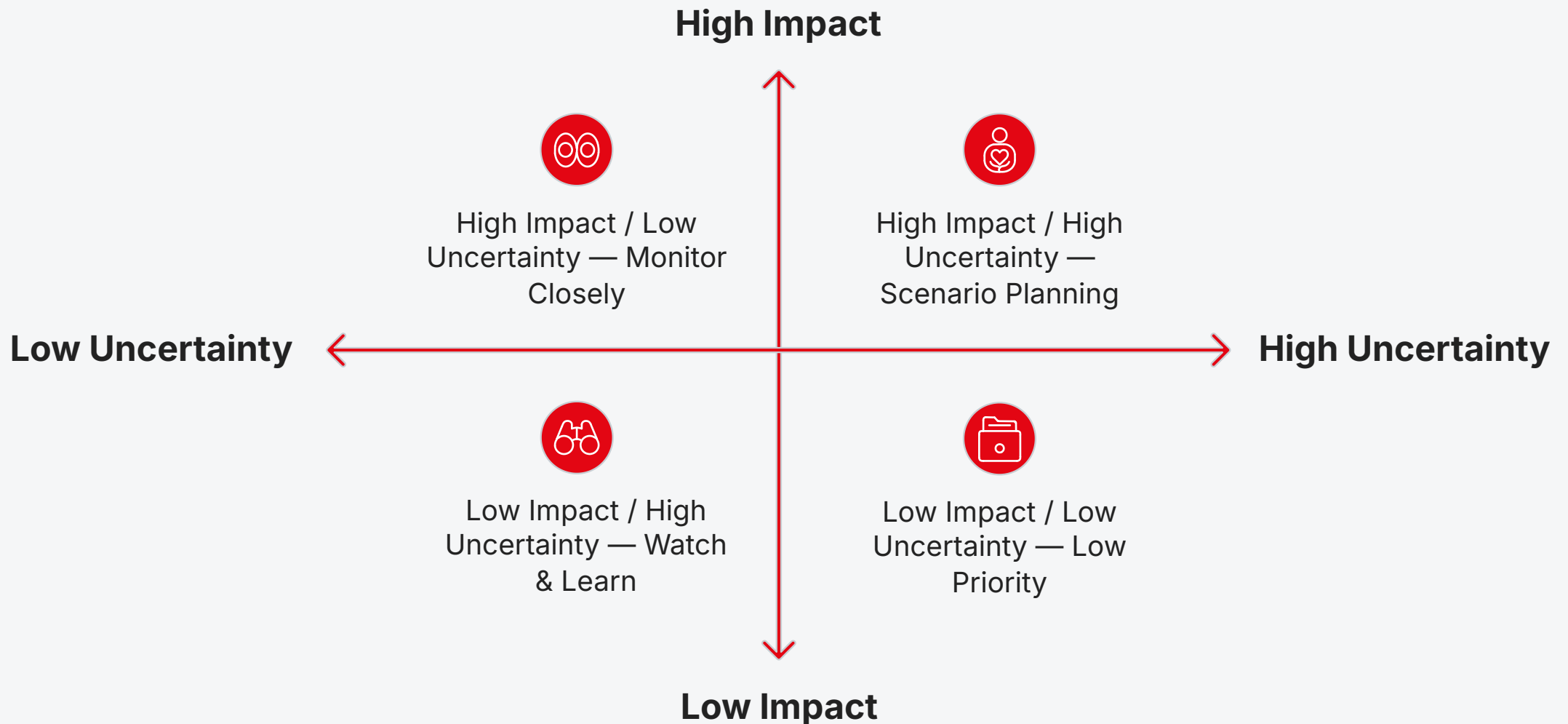


Predictive Analytics

Using statistical models and machine learning to identify patterns in data.

These methods allow organizations to combine qualitative foresight and quantitative analysis.

Trends vs Uncertainties



- Trends: Relatively predictable developments — population growth, urbanization, digitalization
- Uncertainties: High-impact factors with unclear trajectories — political conflicts, technological breakthroughs, climate policy

Systems Thinking in Sustainability

Sustainability challenges involve complex systems composed of many interacting elements.

Energy Systems

Power generation, distribution, and consumption networks

Food Systems

Agriculture, supply chains, and consumption patterns

Transport Systems

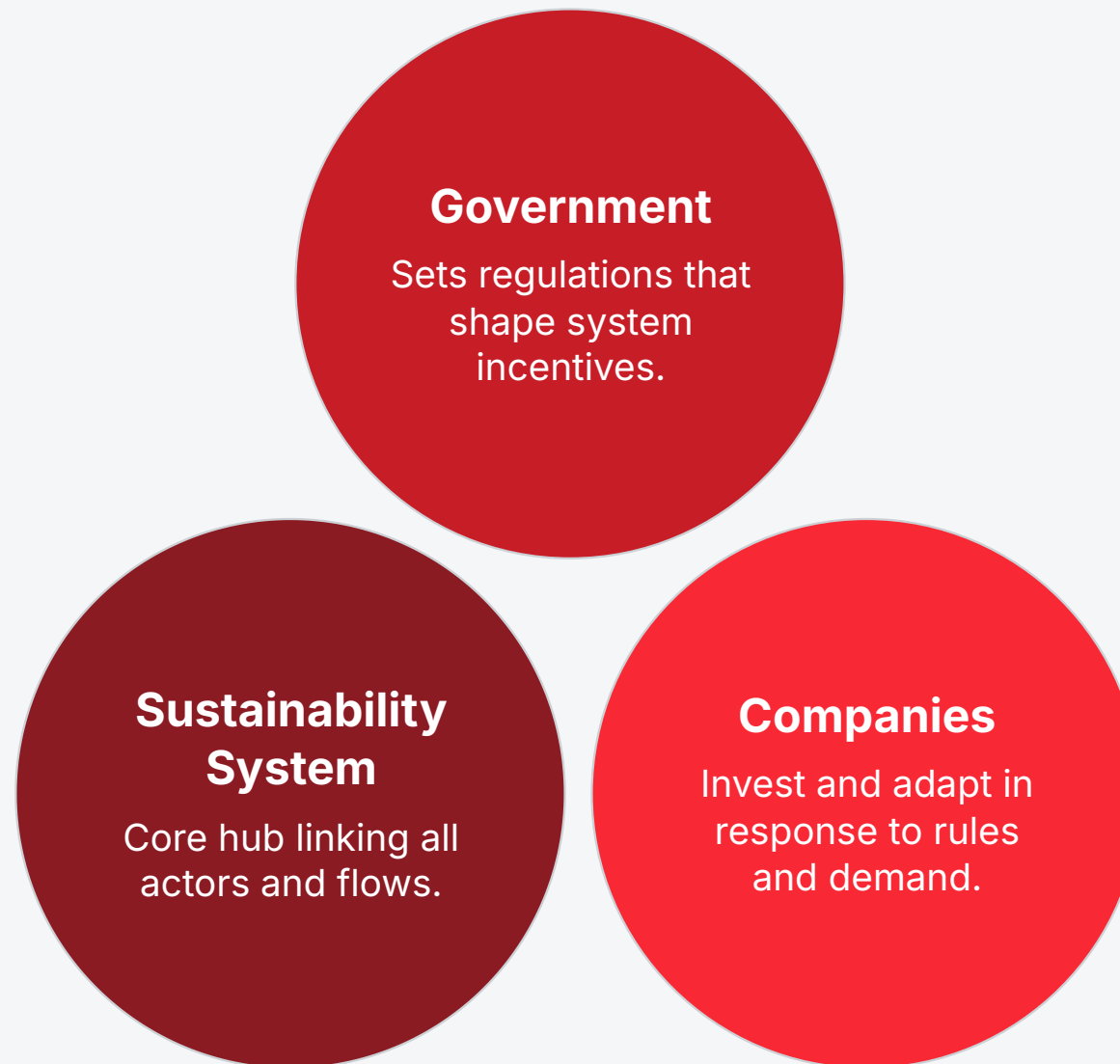
Mobility infrastructure and logistics networks

Global Supply Chains

Cross-border production and distribution flows

Systems thinking helps analysts understand how actions in one part of a system can produce unexpected effects elsewhere — essential for designing effective sustainability strategies.

Sustainability System Map



Sustainability systems involve multiple actors whose decisions influence each other. Government regulations influence corporate investment decisions, which affect technological innovation and consumer behavior. Mapping these relationships helps analysts understand the structure of complex systems.

Feedback Loops



Scenario Planning

One of the most widely used tools in strategic foresight.



Define the Question

Identify the strategic decision or challenge requiring long-term analysis.



Identify Key Uncertainties

Determine the most impactful and uncertain forces shaping the future.



Build Scenarios

Construct multiple detailed narratives representing different possible environments.



Test Strategies

Analyze how current strategies would perform under each scenario.



Identify Robust Actions

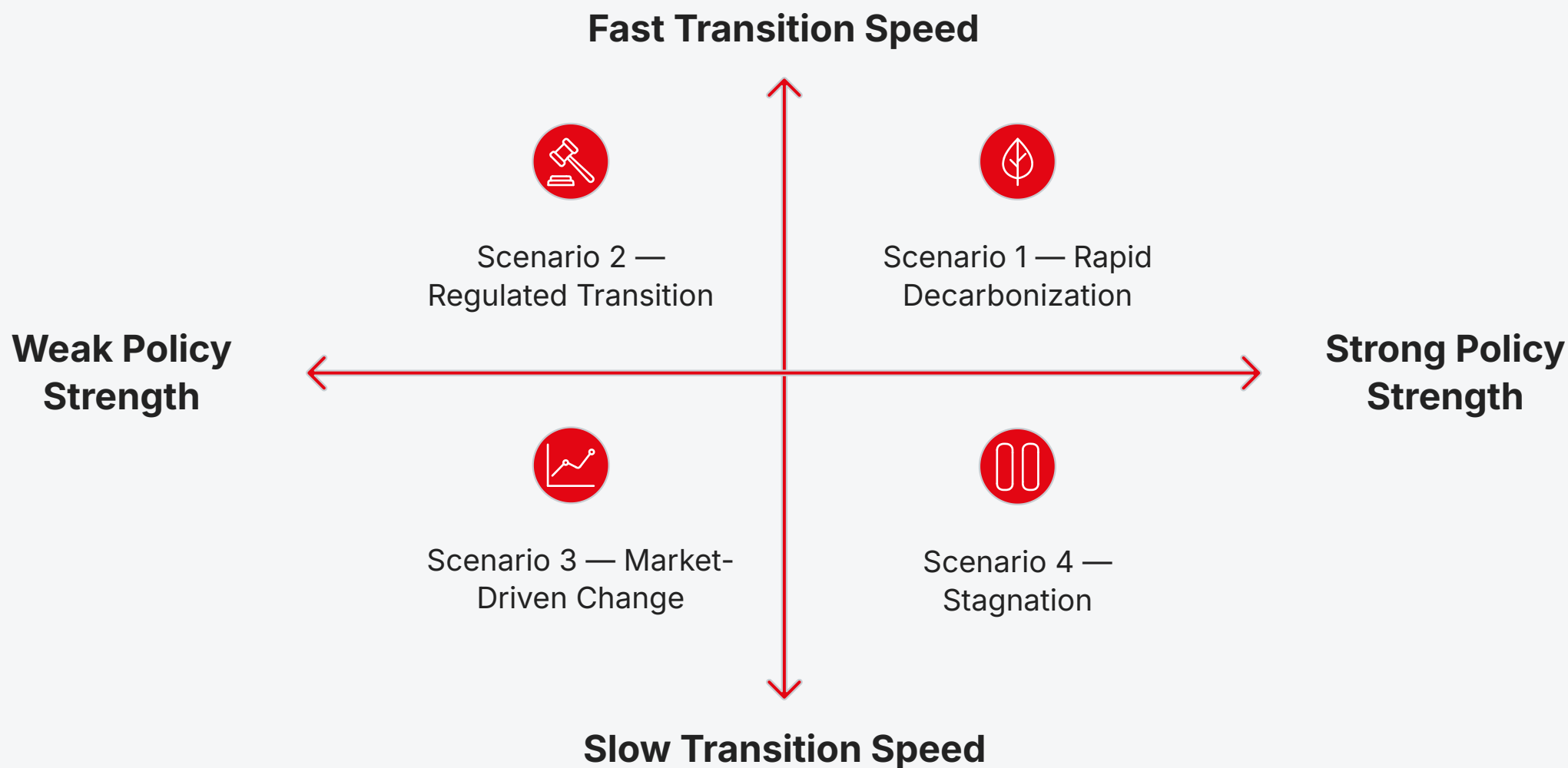
Find strategies that remain effective across multiple futures.

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The goal is not to predict the future — but to be prepared for multiple futures.

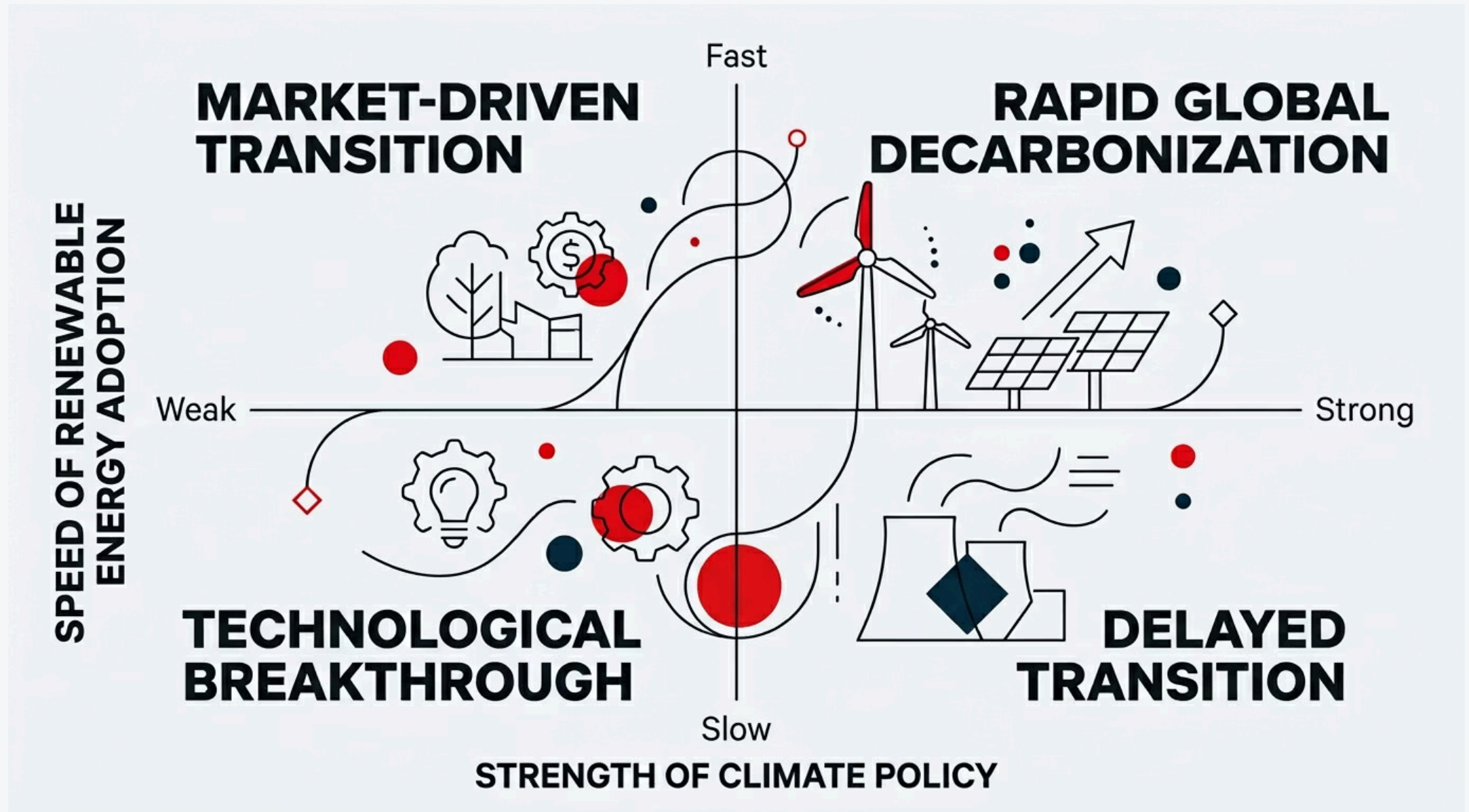
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The Scenario Matrix



Scenario planning begins by identifying two key uncertainties. Combining them produces four distinct scenarios — each representing a different possible future world. Organizations test their strategies against all four.

Example: Energy Transition Scenarios



Energy companies must design strategies that remain viable across all four futures.

Strategic Forecasting in Practice

Many organizations use scenario planning to guide long-term strategy.



Shell

Pioneered energy scenario planning in the 1970s. Shell's scenarios helped the company navigate the 1973 oil crisis and remain a benchmark for corporate foresight.



World Economic Forum

Publishes annual Global Risk Reports exploring future scenarios across geopolitical, environmental, and technological dimensions.



National Governments

Use strategic foresight to design long-term economic, technological, and climate policies. Examples include Finland, Singapore, and the EU.

“Preparing for uncertainty is more valuable than attempting precise predictions.”

Data and AI in Strategic Forecasting

Artificial intelligence increasingly supports strategic forecasting by analyzing large datasets to detect patterns and generate predictions.



Energy Demand Forecasting

AI models predict consumption patterns across regions and seasons to optimize grid management.



Climate Risk Assessment

Machine learning identifies physical and transition risks across geographies and asset classes.



Environmental Monitoring

Satellite imagery analysis detects deforestation, methane emissions, and land-use changes in real time.



Supply Chain Analytics

AI detects disruption risks and optimizes logistics across global networks.

AI complements human strategic thinking by providing powerful analytical capabilities — but interpretation and judgment remain essential.

Example: AI and Climate Monitoring

→ Deforestation Detection

Satellite imagery tracks forest cover loss in near real time across tropical regions.

→ Methane Emissions Tracking

Spectral analysis identifies methane leaks from oil fields, landfills, and agriculture.

→ Land Use Change

AI classifies land cover changes to monitor urban expansion and agricultural conversion.

→ Illegal Fishing Detection

Vessel tracking algorithms identify unauthorized fishing activity in protected waters.

These tools allow governments and organizations to monitor environmental systems in real time — transforming sustainability governance and strategic decision making.

AI-powered monitoring creates new accountability mechanisms and enables evidence-based policy responses at global scale.

From Forecasting to Strategy

Forecasting tools do not automatically generate strategy. Strategic decision making requires combining data analysis with interpretation, judgment, and organizational capabilities.

Data Insights

Quantitative analysis of trends, patterns, and signals from large datasets.



Scenario Analysis

Qualitative exploration of alternative futures and their strategic implications.

Stakeholder Understanding

Knowledge of actor motivations, relationships, and power dynamics.



Organizational Capabilities

Internal resources, competencies, and adaptive capacity.

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The objective is not simply to predict the future — but to design strategies that remain robust across multiple possible futures.

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Preparing for the Capstone Project

Throughout this course you will apply the tools introduced today. Your capstone project mirrors the work of sustainability consultants, policy analysts, and corporate strategy teams.



Analyze a Sustainability System

Select a real-world sustainability challenge and map its key actors, dynamics, and interdependencies.



Collect and Explore Datasets

Gather relevant quantitative and qualitative data to understand current trends and system behavior.



Identify Trends and Uncertainties

Distinguish predictable trends from high-impact uncertainties that will shape the future.



Develop Alternative Scenarios

Construct two to four plausible future scenarios using the scenario matrix method.



Propose Strategic Recommendations

Design robust strategies that address sustainability challenges across multiple possible futures.

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The goal is to develop strategies that address both complex sustainability challenges and uncertain futures.

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